Vivi Lazo

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Education

- **University of Central Florida** Orlando, FL
  Bachelor's of Science in Computer Science GPA: 3.55 *05/2025*
  - Computer Science I & II, Object-Oriented Programming, Database Systems, Statistical Methods I & II, Software Engineering, Processes of Object-Oriented Software Development , Processes of Object-Oriented Software Development,
- **University of Central Florida** Orlando, FL
  Master's of Science in Data Analytics GPA: 3.917 *05/2027*
  - Network Science, Statistical Methods, Data Mining I & II, Machine Learning, Parallel & Distributed Database Systems, Computational Analysis of Social Complexity, Parallel Cloud Computation

Technical Skills

- Languages/Frameworks: Python, Pandas, Numpy, Scikit-learn, Ollama, Huggingface Transformers, Flask, R, Julia, PHP, MongoDB, Postgres/MySQL, Excel
- Software & Operating Systems: Anaconda, Docker, Linux, Windows, LM Studio, Github, Figma

Professional Experience

- **Mixed Emerging Technology Integration Lab (METIL)** Orlando, FL
  University of Central Florida *February 2024 - Present*
  - Designed three data pipelines that integrates local LLM technologies to handle sensitive documentation in three distinct research projects.
  - Managed a team of 5 student interns in a Unity-based Glovebox simulation project sponsored by the Department of Energy.
  - Implemented semantic search algorithms to optimize company-specific queries from a vector database for over 150 employees.

Projects

- **Code Repair with LLMs** Orlando, FL
  Senior Design @ UCF *September 2024 - May 2025*
  - Designed a four-step Fine-Tuning pipeline to identify and debug vulnerability patterns in code repositories containing 10,000 LOC using HuggingFace Transformers and Gradio.
  - Experimented with training code generation models on +5,000 CVEs to perform code repair on C, Java, and Python programs using LoRA.
  - Evaluated the benefits of implementing Retrieval-based Augmented Generation (RAG) and Fine-Tuning methods simultaneously to enhance model response quality by using LangChain and Ollama.
- **Portable AI Knowledgebase** Orlando, FL
  DevOps @ METIL *September 2024 - February 2025*
  - Engineered a locally deployable system with RAG and fine-tuning capabilities on low-resource devices using Python and Gradio.
  - Leveraged FAISS to embed 50 task-specific documents in a local vector database and deployed it using a Docker image.
  - Assessed and fine-tuned the performance of the Meta-Llama 3.1 8B-Instruct model to produce concise summaries via utilizing LM Studio.
- **Human Performance Test Facility Project** Orlando, FL
  DevOps @ METIL *April 2024 - August 2024*
  - Developed a Python script to calculate performance metrics of 30 experiment participants by using Pandas for data manipulation and analysis.
  - Processed 116 experiment tasks by using Excel workbooks and CSV files as the input/output mediums.
  - Produced a dataset with 100 performance metrics per participant to be used for further statistical analysis.